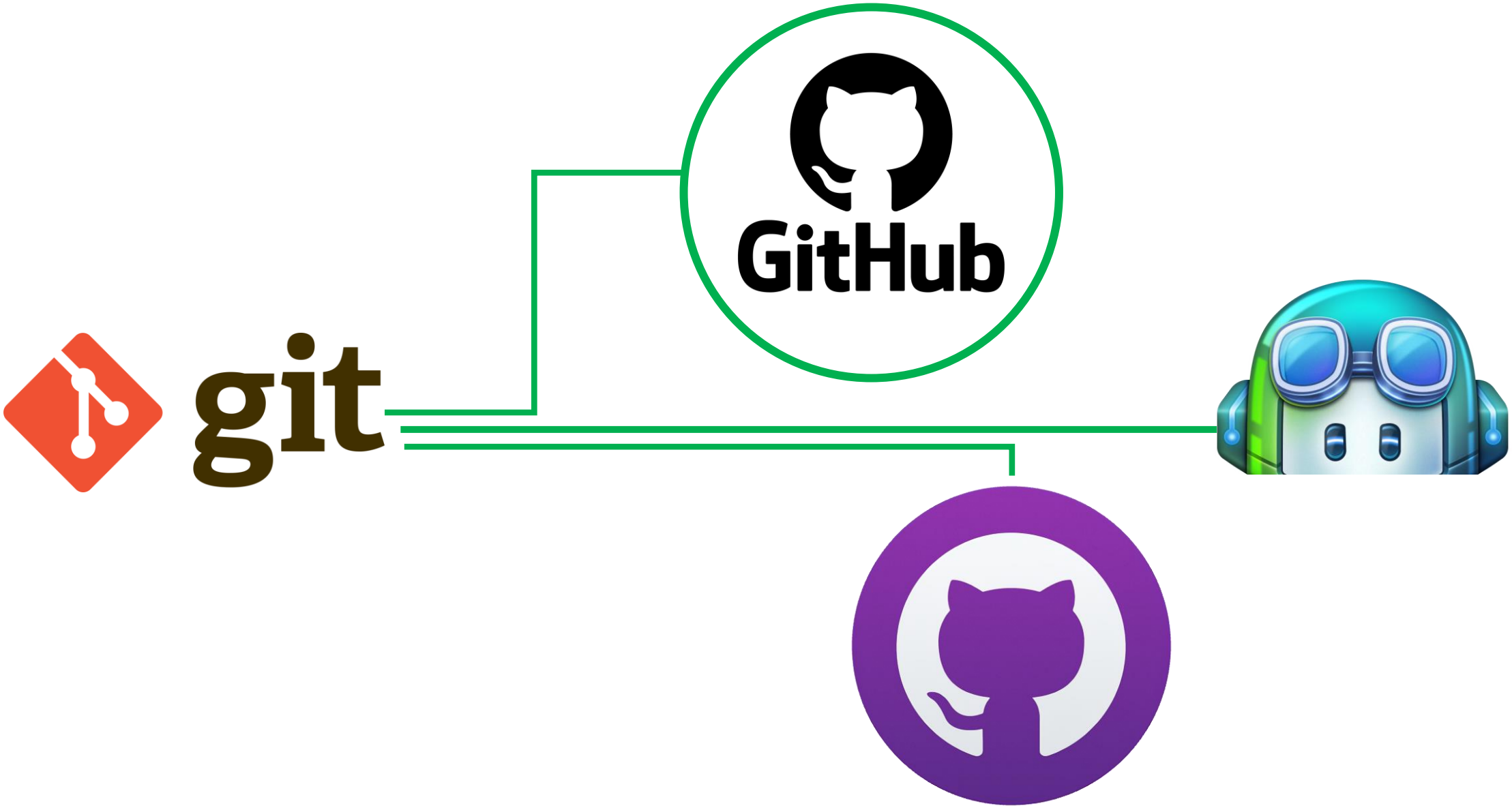
실무에서 활용하는 Git 특강



jeongeunswd@gmail.com

Git이란 무엇인가?

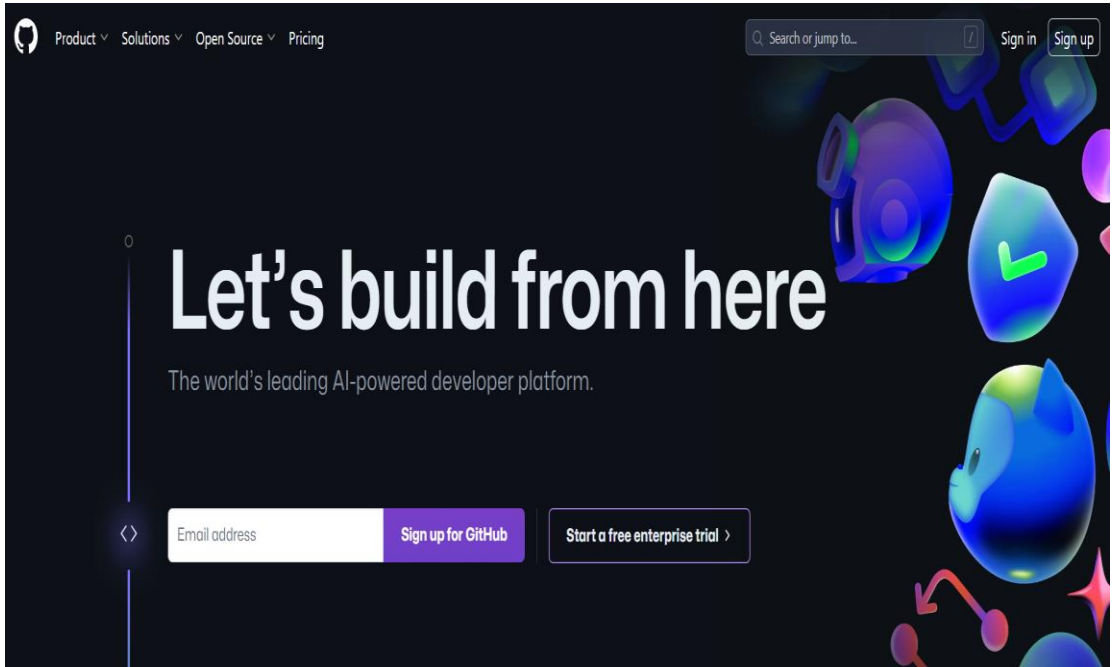
- 분산 버전 관리 시스템.
- 소스 코드의 변경 사항을 추적하고 여러 개발자들이 동시에 작업할 수 있게 도와 줌
- 주요 기능 : 소스 코드의 버전 관리, 브랜치(branch) 관리, 내용 병합, 충돌 해결 등

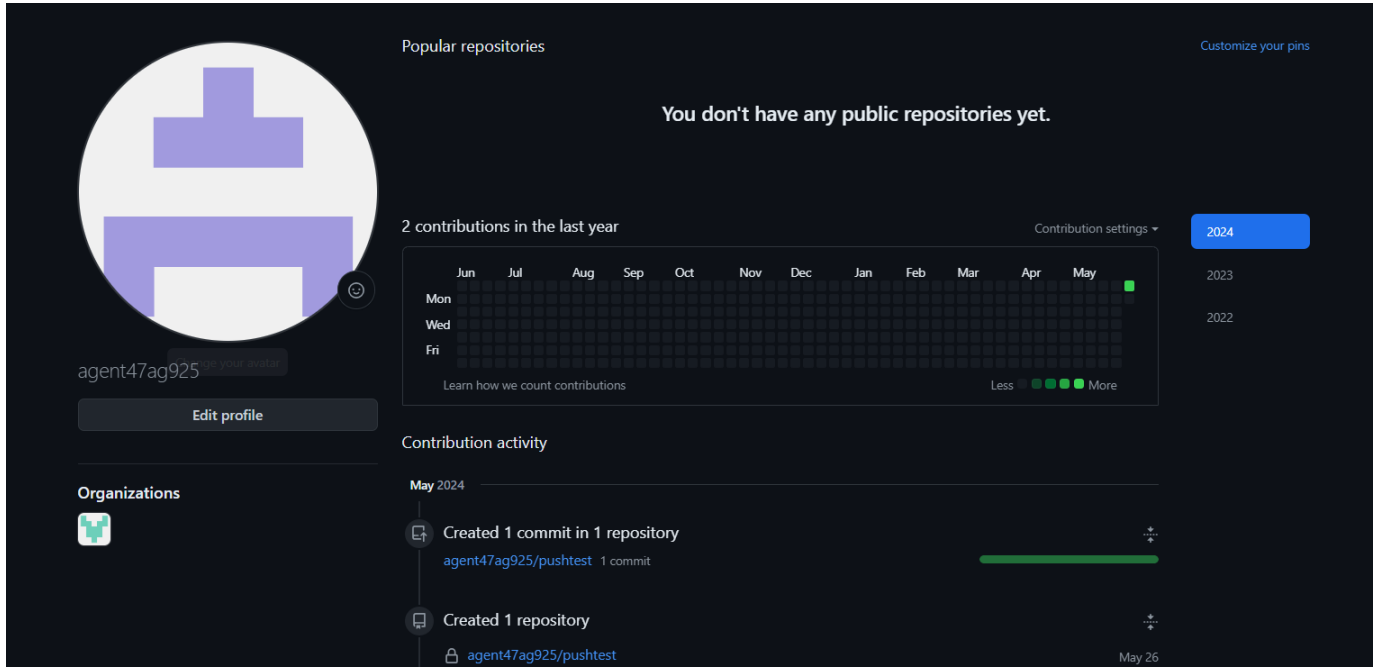
Git is a 무료 오픈 소스 distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is 배우기 쉽고 and has a 엄청 가볍고 빠르게 동작함 performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like 가벼운 로컬 브랜칭, convenient staging areas, and 편리하게 여러 개의 워크플로우를 만들 수 있음

GitHub란 무엇인가?

- https://github.com/
- GitHub는 Git을 기반으로 하는 웹 호스팅 서비스
- 소스 코드 저장소를 온라인에서 관리할 수 있게 해 줌
- 주요 기능 : 원격 저장소 제공, 협업 관리, 이슈 추적, 프로젝트 관리, Actions





GitHub와 비슷한 서비스는?

- Git 시스템을 기반으로 온라인 저장소와 소스코드 협업을 제공하되, 특색 있는 서비스들이 많음

GitLab	Bitbucket	Source Forge	AWS Code Commit	Azure Repos
• 온라인 저장소 • CI/CD 파이프라인 • 이슈 트래킹 • 코드 리뷰	• 온라인 저장소 • CI/CD 파이프라인	• 온라인 저장소 • 프로젝트 호스팅 • 이슈 트래킹	• 온라인 저장소 • AWS 다른 서비스와 통합이 용이	• 온라인 저장소 • CI/CD 파이프라인

GitLab

SOURCEFORGE

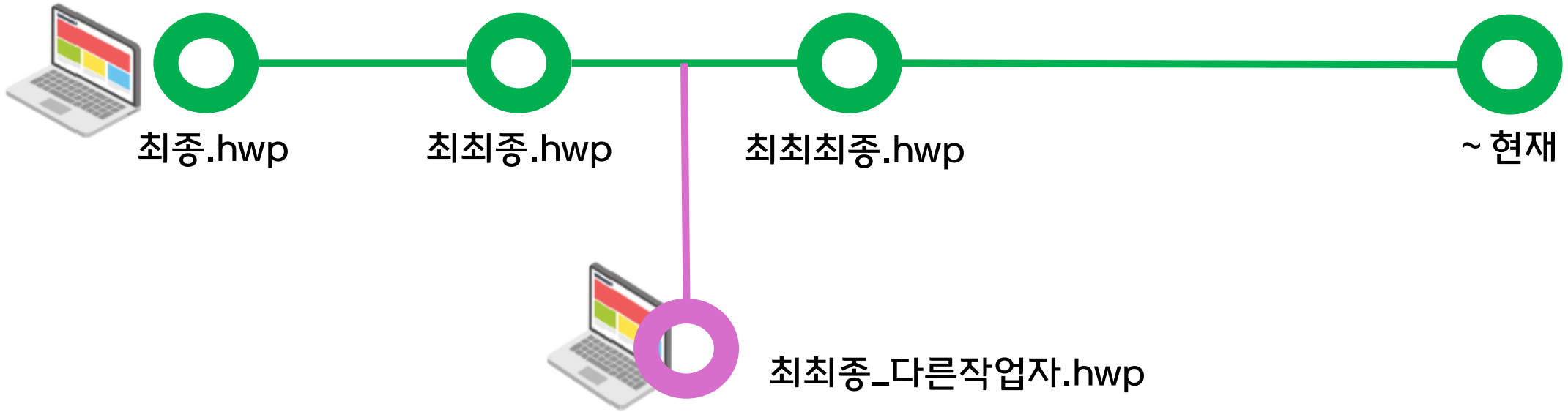
Bitbucket

AWS CodeCommit

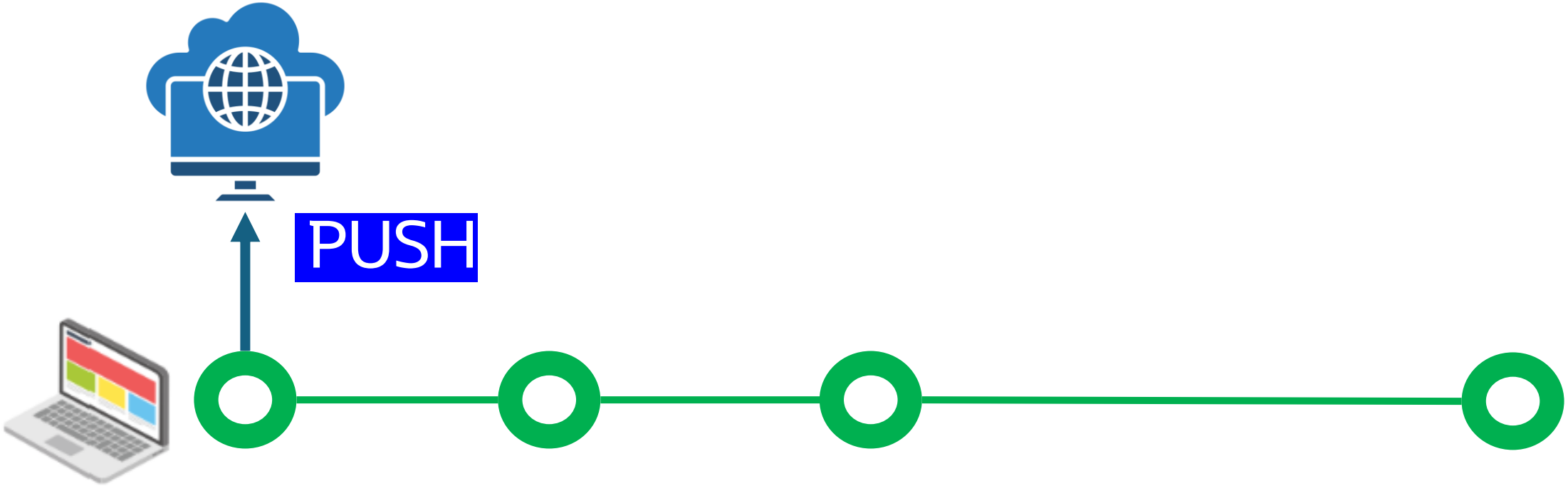
Azure Repos

Git의 개념

- **저장소(repo)** : 프로젝트 파일과 변경 이력을 관리하는 디렉토리. 로컬저장소와 원격저장소로 나누어짐
- **브랜치**(branch) : 저장소에서 작업하는 워크플로우. 기본 브랜치는 main이나 master로 불리며,
  필요하면 별도의 브랜치를 만들어 작업할 수 있음
- **커밋(commit)** : 파일의 변경 사항을 저장소에 기록하는 작업. 특정 시점의 코드 상태를 저장하여,
  추후에 돌아갈 수(Rollback) 있음
- **병합(merge)** : 2명 이상, 혹은 원격 저장소와 로컬 저장소의 같은 파일을 하나의 내용으로 합치는 작업.

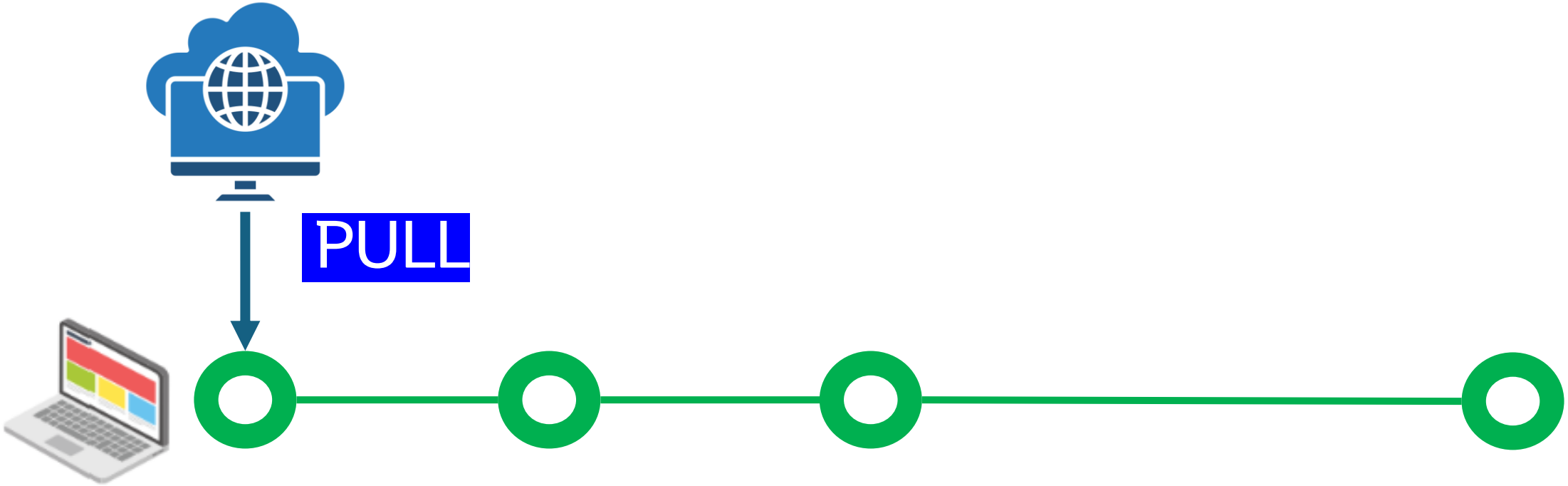


Git의 주요 행동 -push



- git init
- git add .
- git commit -m “커밋 메세지”
- git push origin main(master)

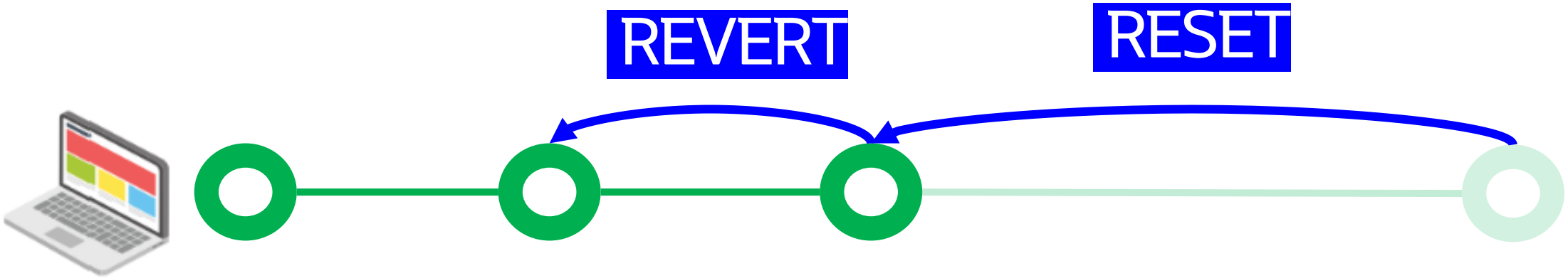
Git의 주요 행동 -pull(pull request)



- git pull origin main(master)
  - git pull --allow-unrelated-histories origin main

- git fetch origin main(master)
- git merge origin main(master)

Git의 주요 행동 – 시점 변경

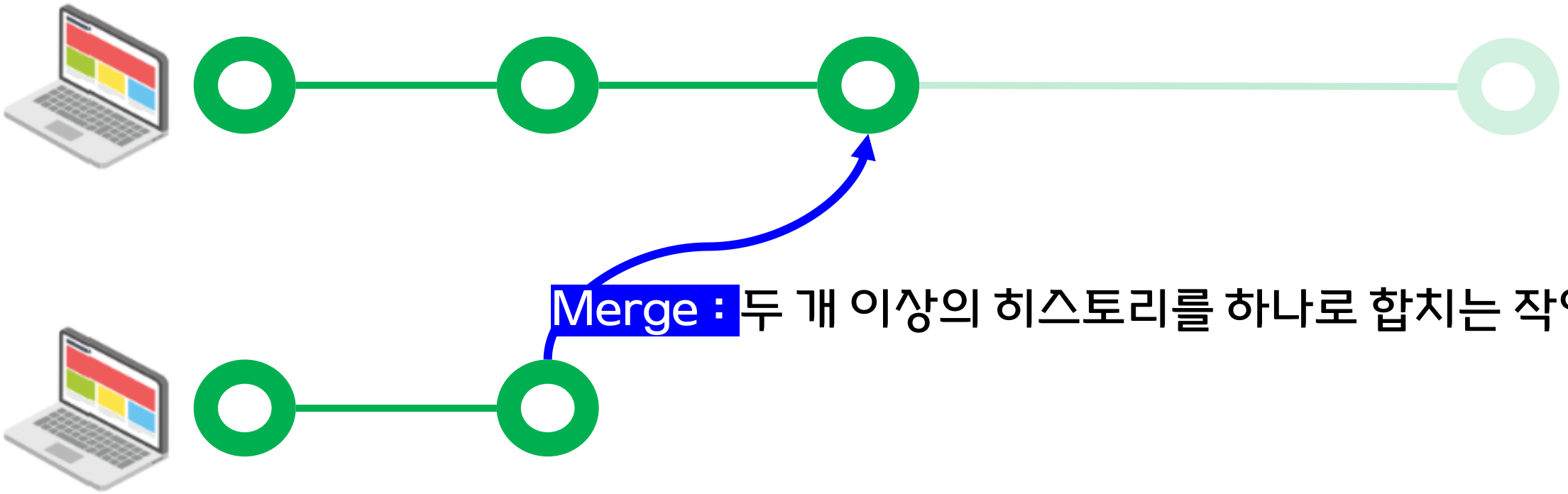


- git revert <Commit ID>

- git reset <Commit ID>
- Soft : 디렉토리와 기록을 남김
- Hard : 디렉토리까지 삭제

Git의 주요 행동 – 시점 변경

- git merge <Commit ID>



Merge : 두 개 이상의 히스토리를 하나로 합치는 작업

GitHub 계정 만들기 실습

https://github.com/

Git 계정 세팅하기

- 로그인 방법 : github의 계정으로 로그인하여 저장소에 접근
- 액세스 토큰 만들기 : github의 설정에서 액세스 토큰 만들어 저장소에 접근
- SSH key 방법 : ssh 키를 만들고, 기기에서 이를 인증하여 저장소에 접근

Git Bash 터미널 설치

Downloads

macOS Windows

Linux/Unix

Older releases are available and the Git source repository is on GitHub.

Download for Windows

Click here to download the latest (**2.45.1**) **64-bit** version of **Git for Windows**. This is the most recent maintained build. It was released **12 days ago**, on 2024-05-14.

Other Git for Windows downloads

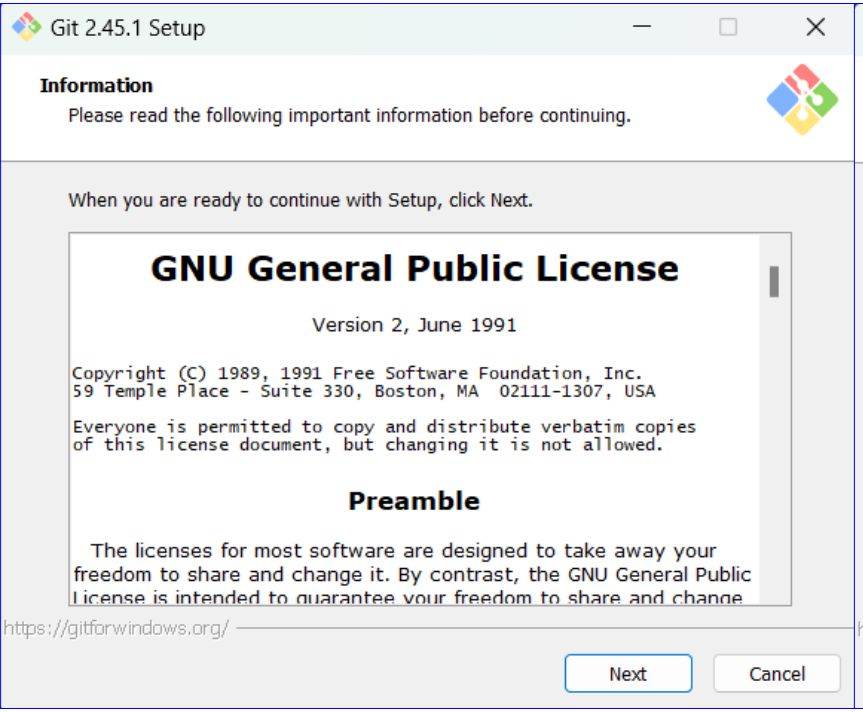
Standalone Installer

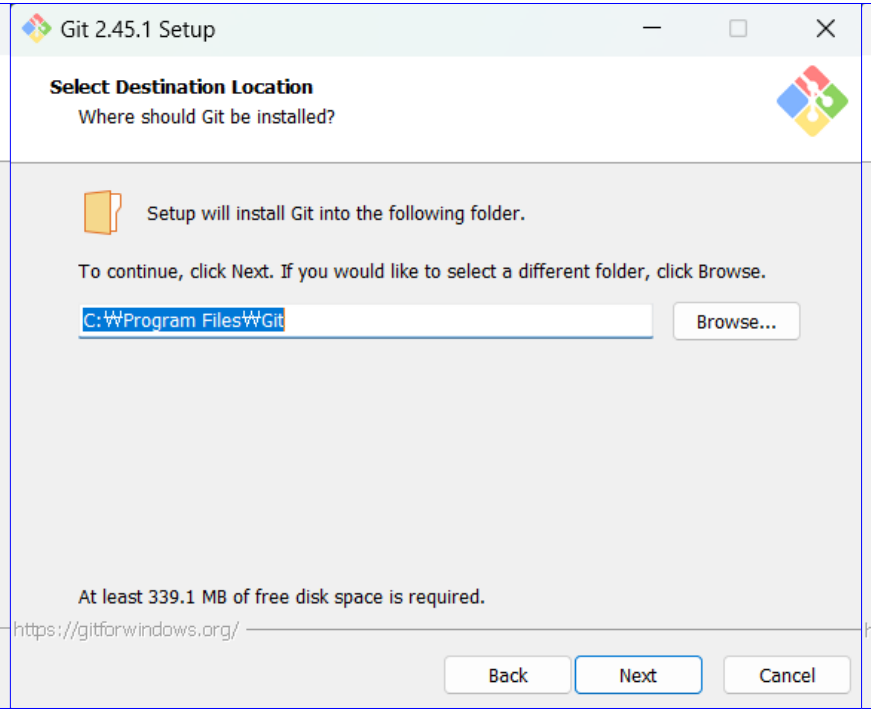
32-bit Git for Windows Setup.

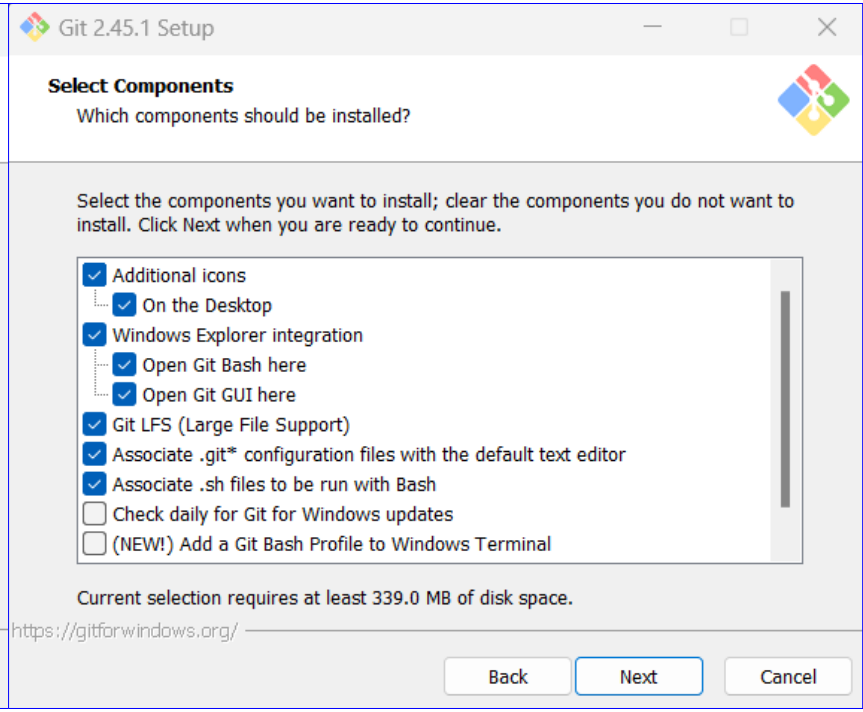
64-bit Git for Windows Setup.

https://git-scm.com/downloads

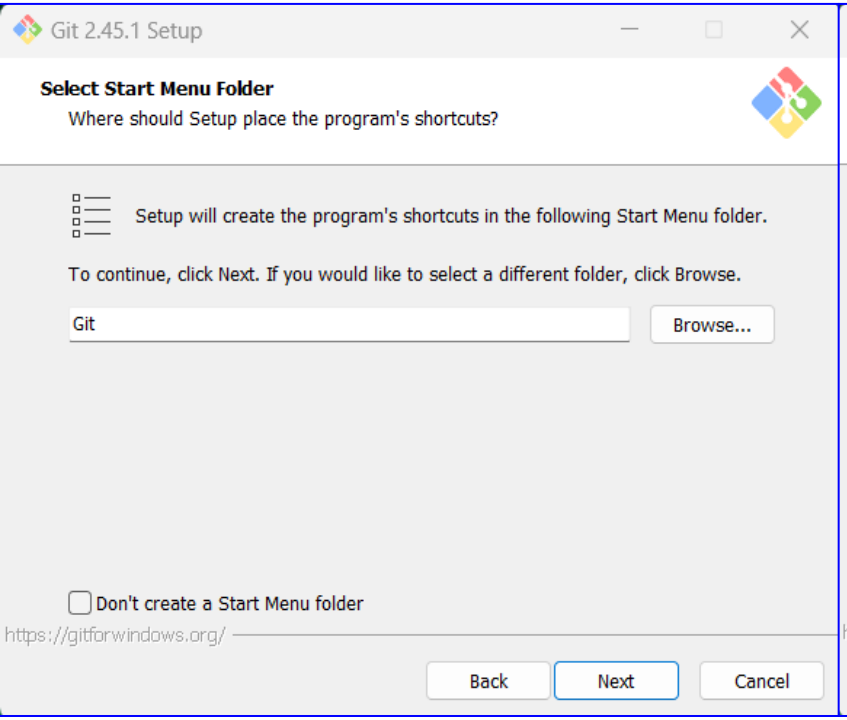
Git Bash 터미널 설치

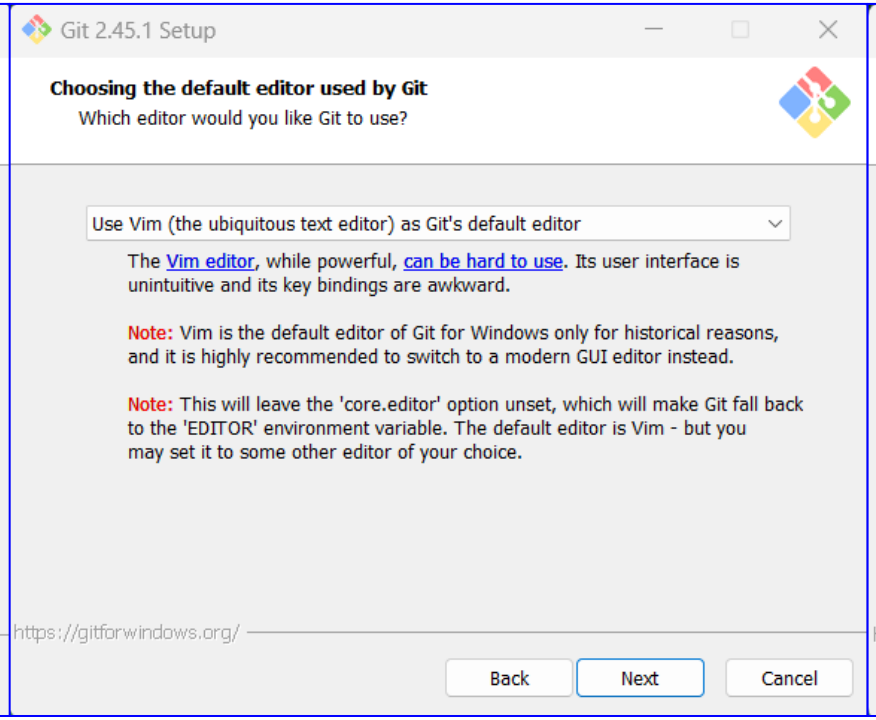


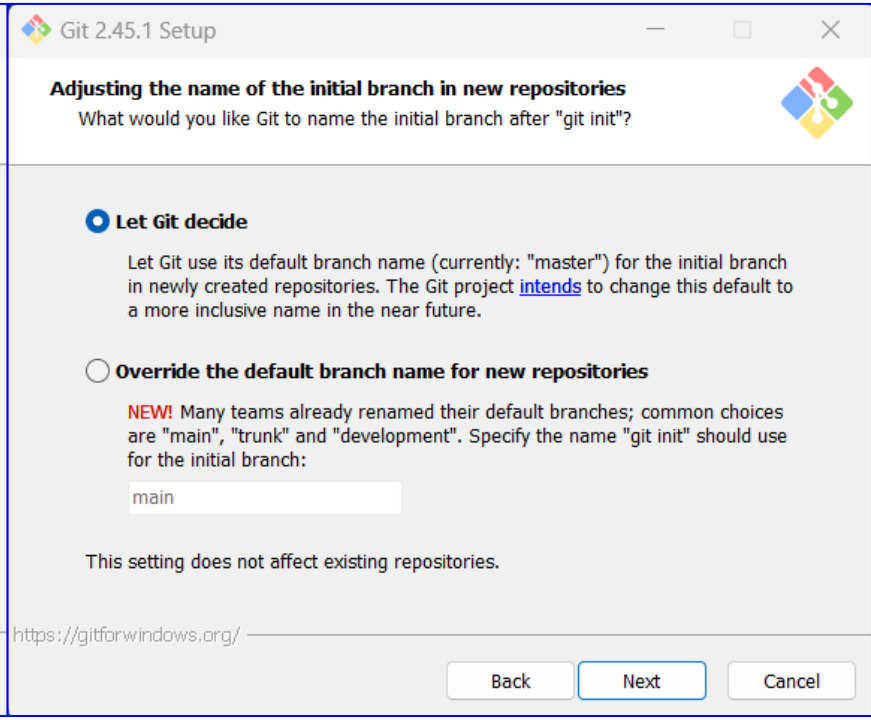




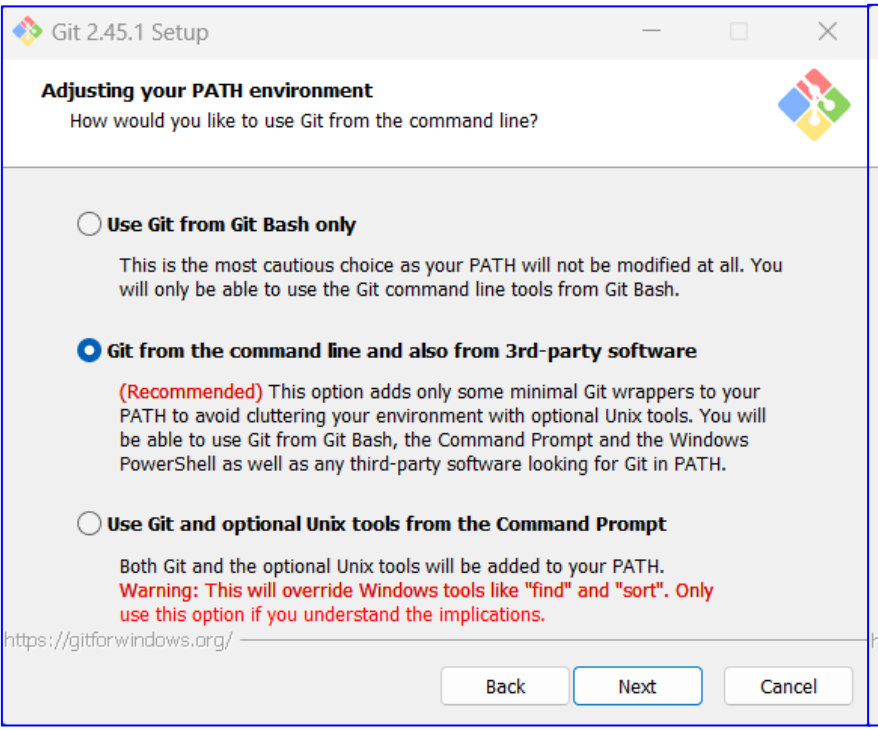
Git Bash 터미널 설치

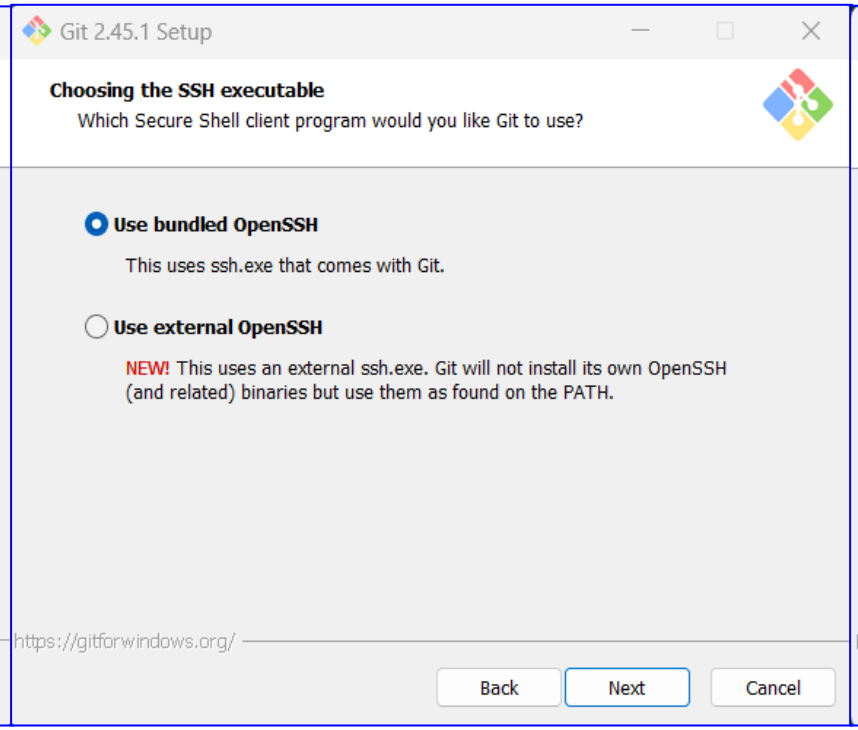


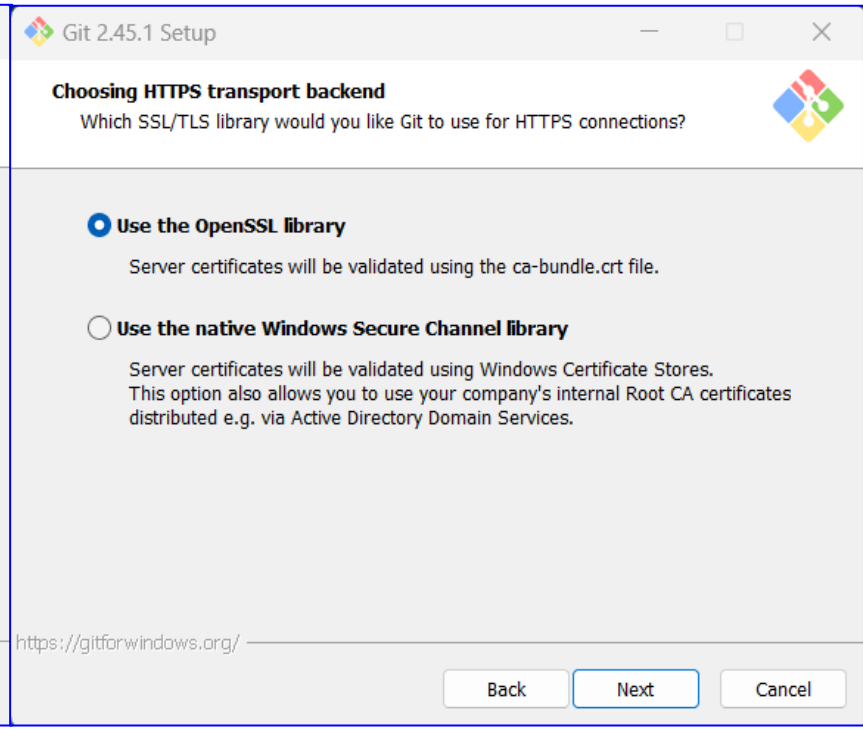




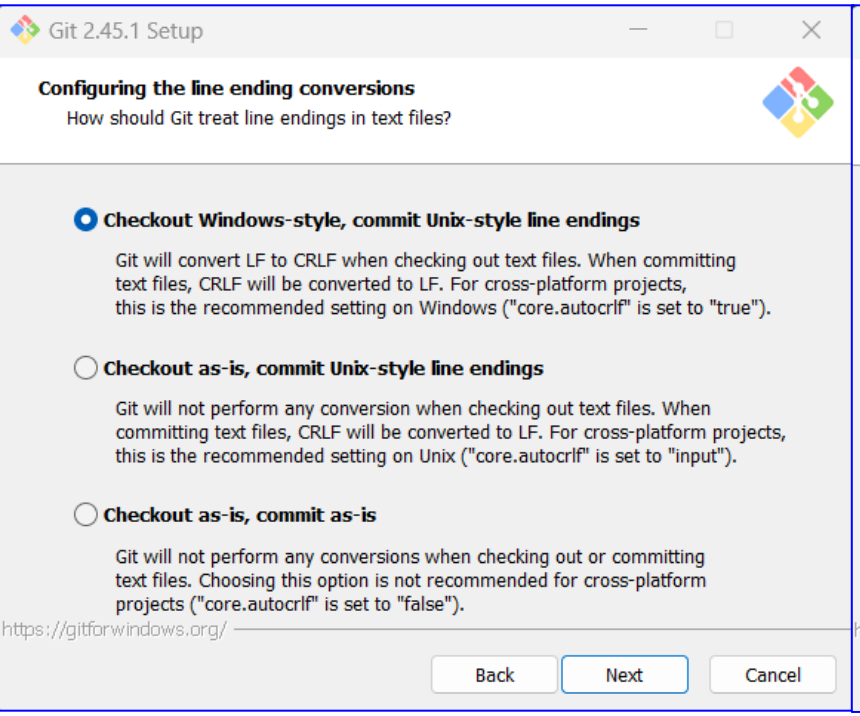
Git Bash 터미널 설치

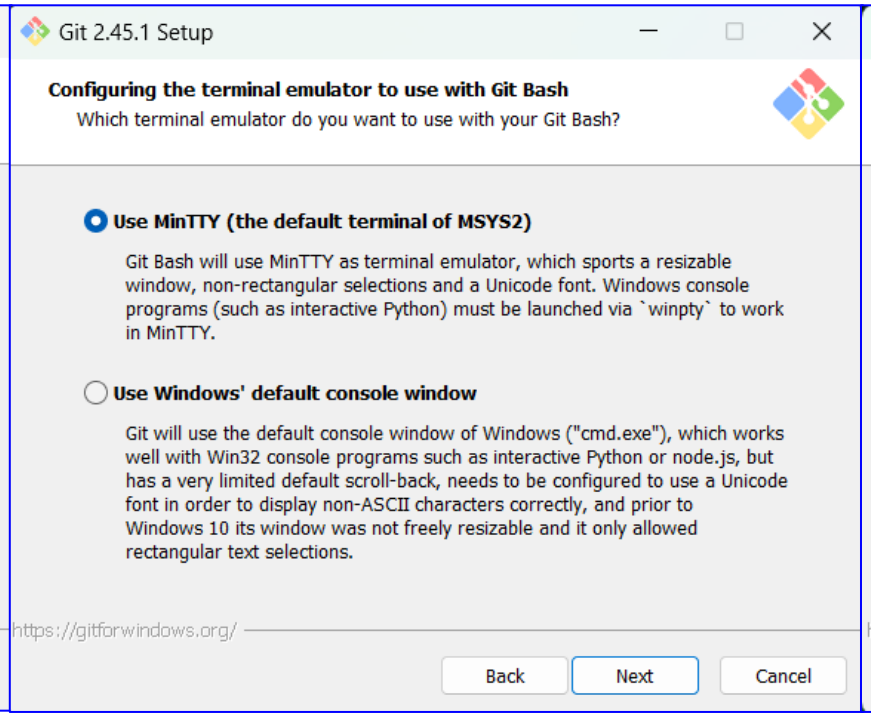


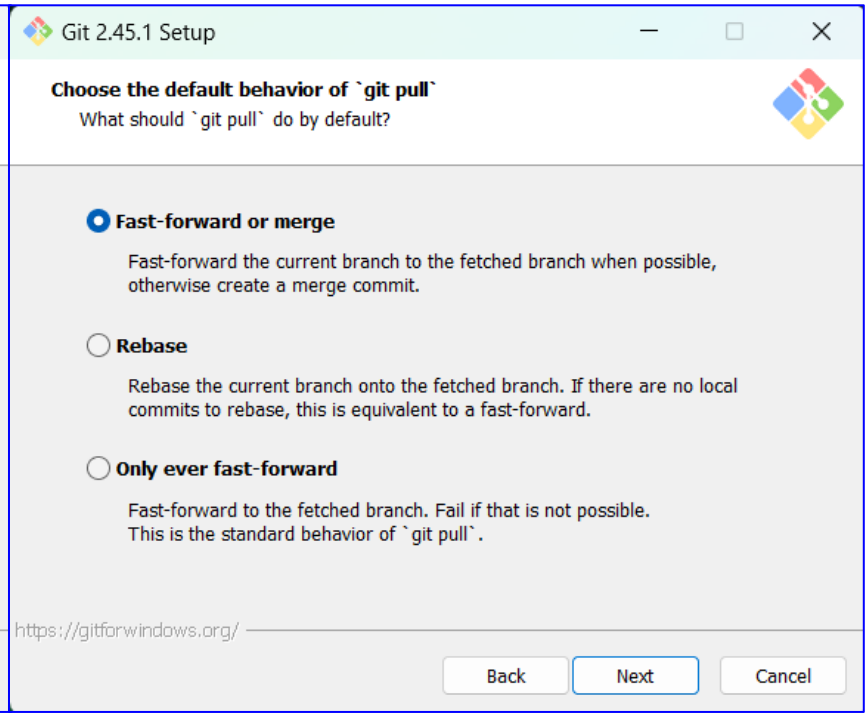




Git Bash 터미널 설치







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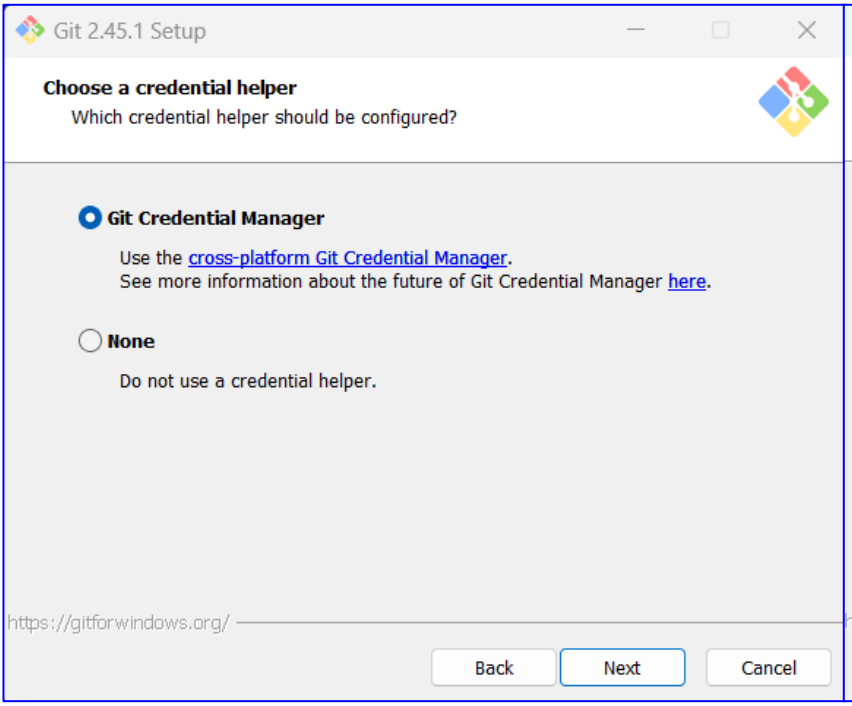

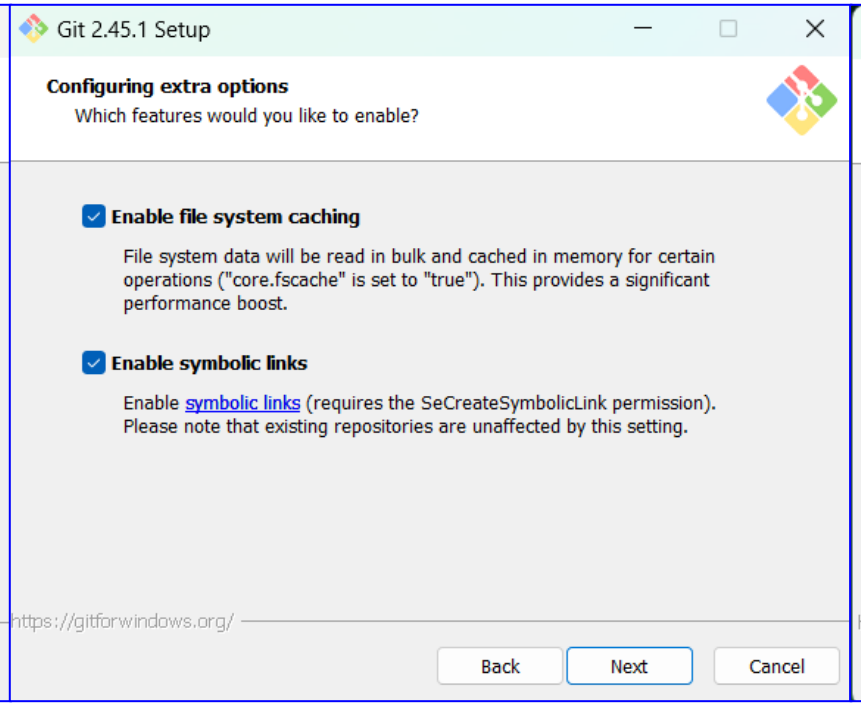

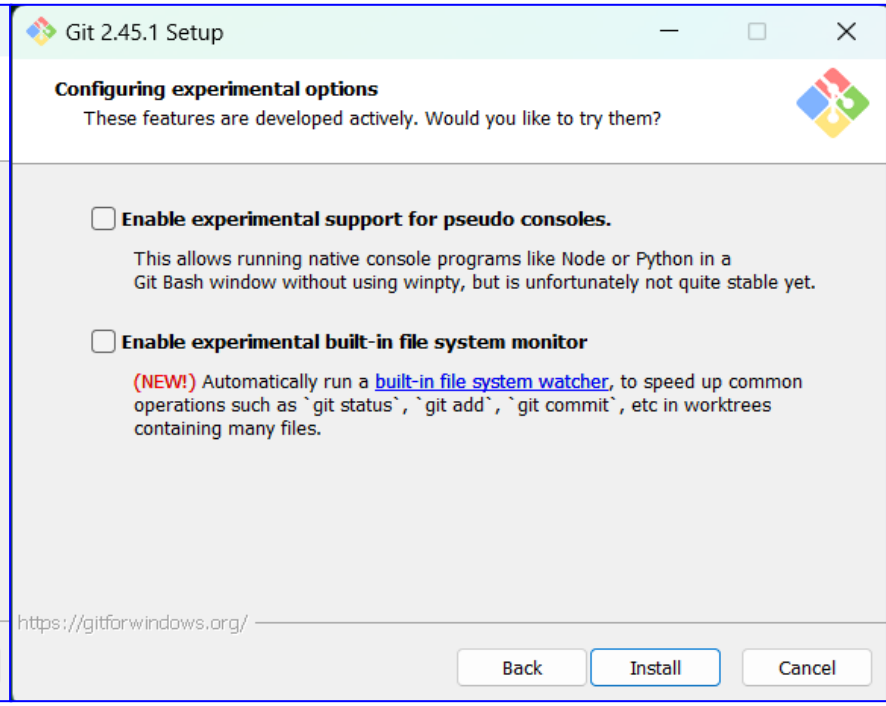

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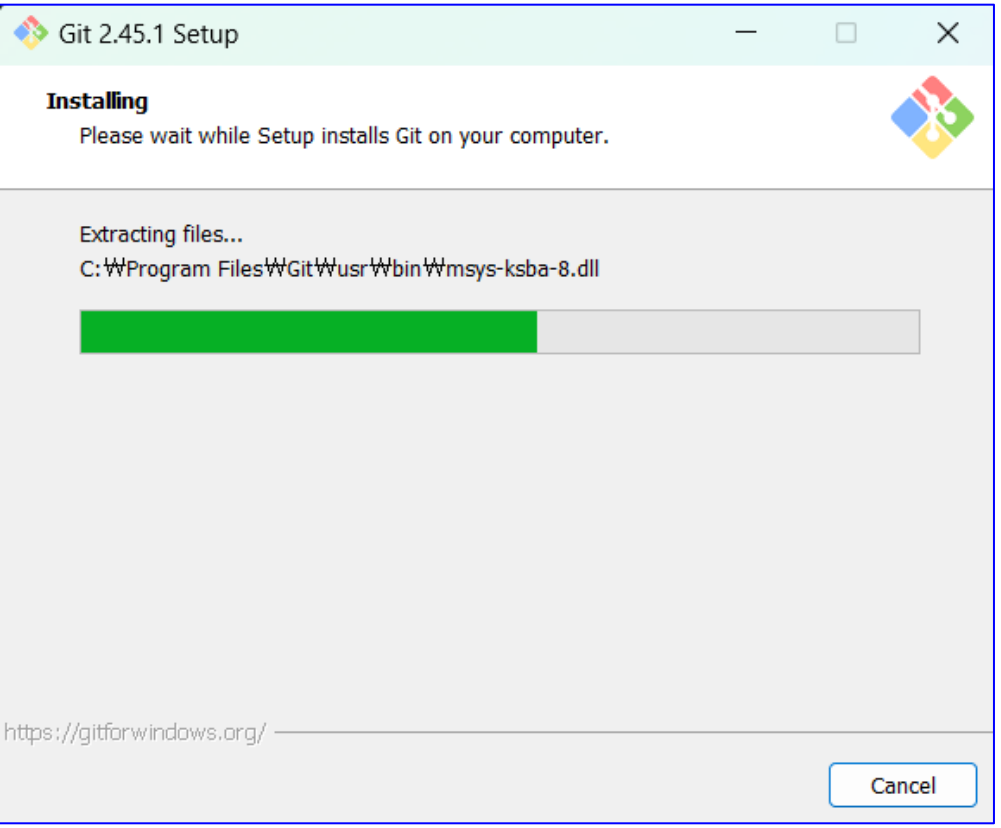
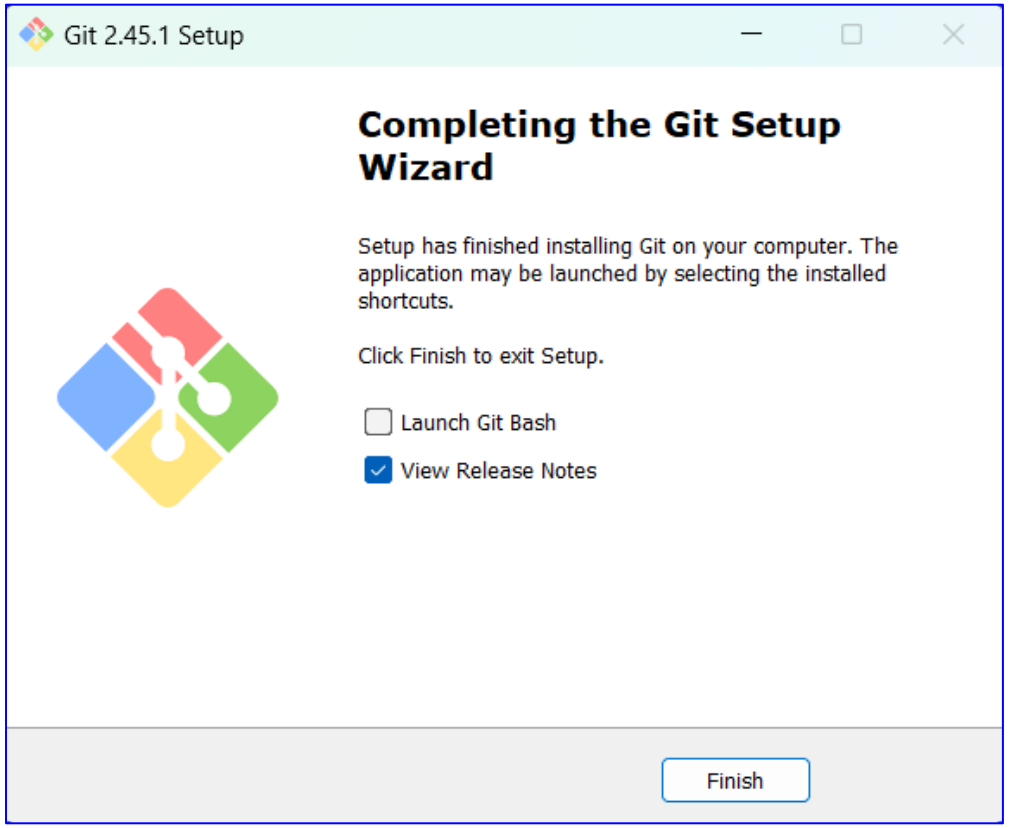
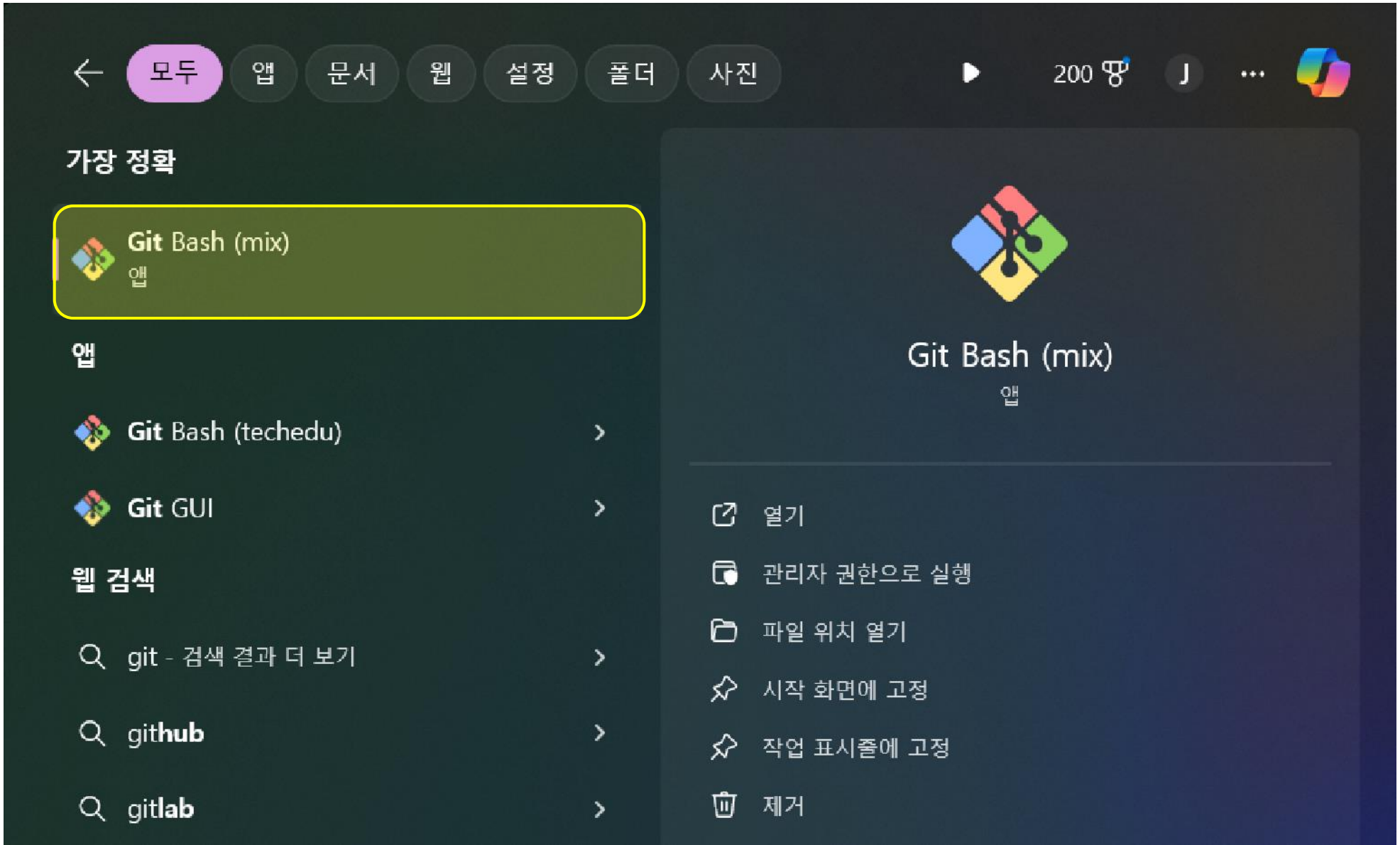




Git Bash 터미널 설치



GitHub 레포 만들기

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? Import a repository.

Required fields are marked with an asterisk (*).

Owner * agent47ag925 / **Repository name ***

레포지토리 이름

Great repository names are short and memorable. Need inspiration? How about ideal-computing-machine ?

Description (optional)

Public
Anyone on the internet can see this repository. You choose who can commit.

퍼블릭(타인이 볼 수 있음)

Private
You choose who can see and commit to this repository.

프라이빗(잠금, 소수만)

Initialize this repository with:

Add a README file
This is where you can write a long description for your project. Learn more about READMEs.

README 설명파일

Add .gitignore

.gitignore template: None

.gitignore (깃 추적 무시 파일)

Choose which files not to track from a list of templates. Learn more about ignoring files.

Choose a license

License: None

라이선스 설정

A license tells others what they can and can't do with your code. Learn more about licenses.

GitHub 레포 연결하기

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest
$ git init
Initialized empty Git repository in C:/Users/jeong/Documents/GitTest/pushtest/.git/

jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git remote set-url main https://agent47ag925@github.com/agent47ag925/pushtest.git
```

잘못되어 지울 때 : git remote remove origin

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git remote -v
origin  https://github.com/agent47ag925/pushtest.git (fetch)
origin  https://github.com/agent47ag925/pushtest.git (push)
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ echo "NEW REPOSITORY TEST" >> README.text
```

GitHub 레포 연결하기

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ ls
README.text
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.text

nothing added to commit but untracked files present (use "git add" to track)

jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git add .
warning: in the working copy of 'README.text', LF will be replaced by CRLF the next time Git
touches it
```

GitHub 레포 연결하기

```
Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git config --global user.email 47ag925park@gmail.com

jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git config --global user.name agent47ag925
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git config --list
user.email=47ag925park@gmail.com
user.name=agent47ag925
```

GitHub 레포 연결하기

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git commit -m "+First commit"
[master (root-commit) 62447ac] +First commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.text
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ git push origin master
remote: Repository not found.
fatal: repository 'https://github.com/agent47ag925/pushtest.git/' not found
```

SSH KEY 생성 인증

GitHub 레포 연결하기

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ ssh-keygen -t ed25519 -C "47ag925park@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/jeong/.ssh/id_ed25519):
Created directory '/c/Users/jeong/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/jeong/.ssh/id_ed25519
Your public key has been saved in /c/Users/jeong/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:xziU7FR88QUid7O4vTAmVHhONHtS5U+IQig6pWNCkPg 47ag925park@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|oo       o+oB.=oo|
|o .   o..+o++X * |
| o   + .= o== = o|
|  E *  + + ..= ..|
|   o o  S + = . .|
|         o o o . |
|              .  |
|                 |
|                 |
+----[SHA256]-----+
```

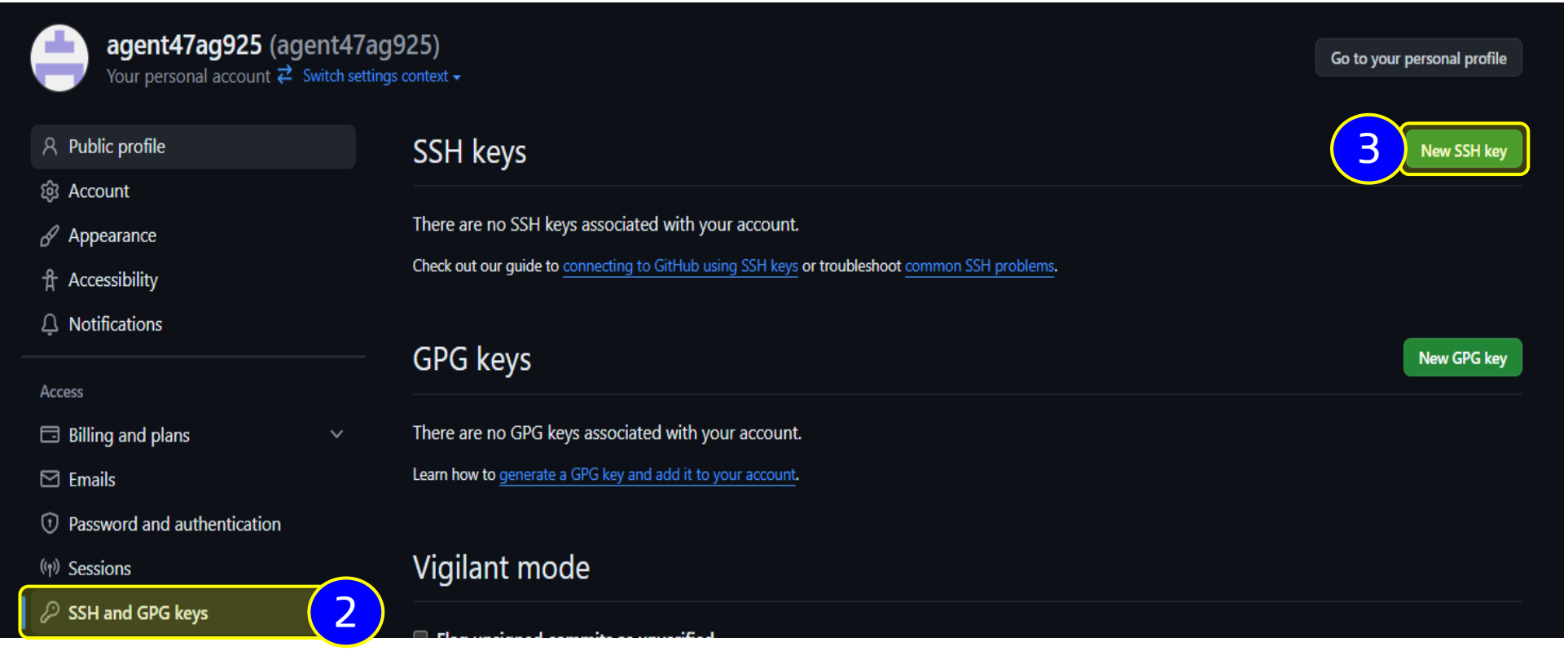
ssh-keygen -t ed25519 -C “이메일”

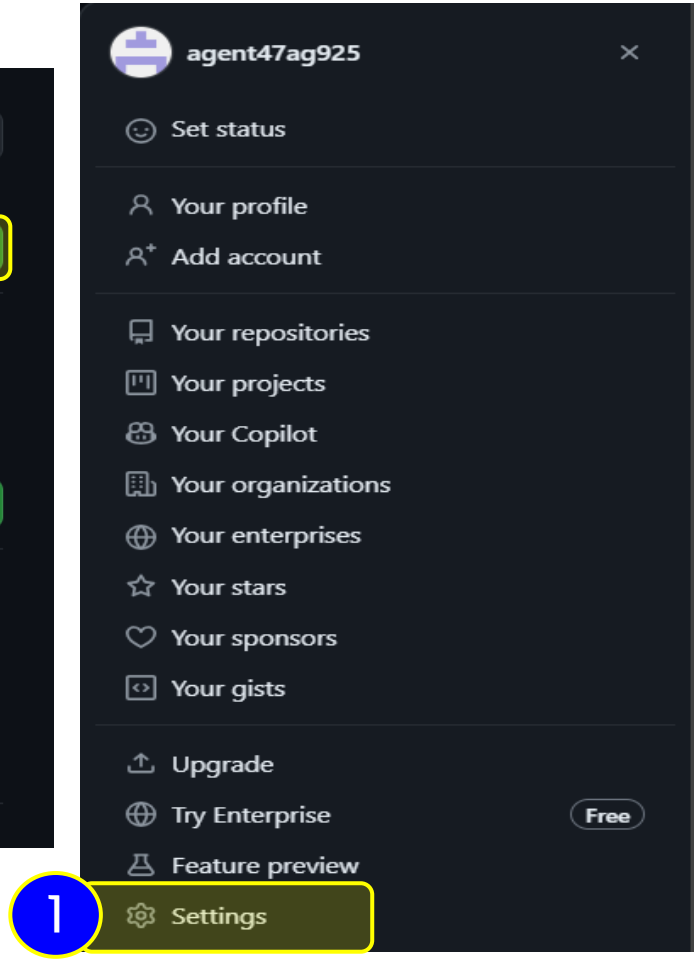
ssh-keygen -t rsa -b 4096 -C “이메일”

GitHub 레포 연결하기 - 내 깃헙 settings에 연결하기

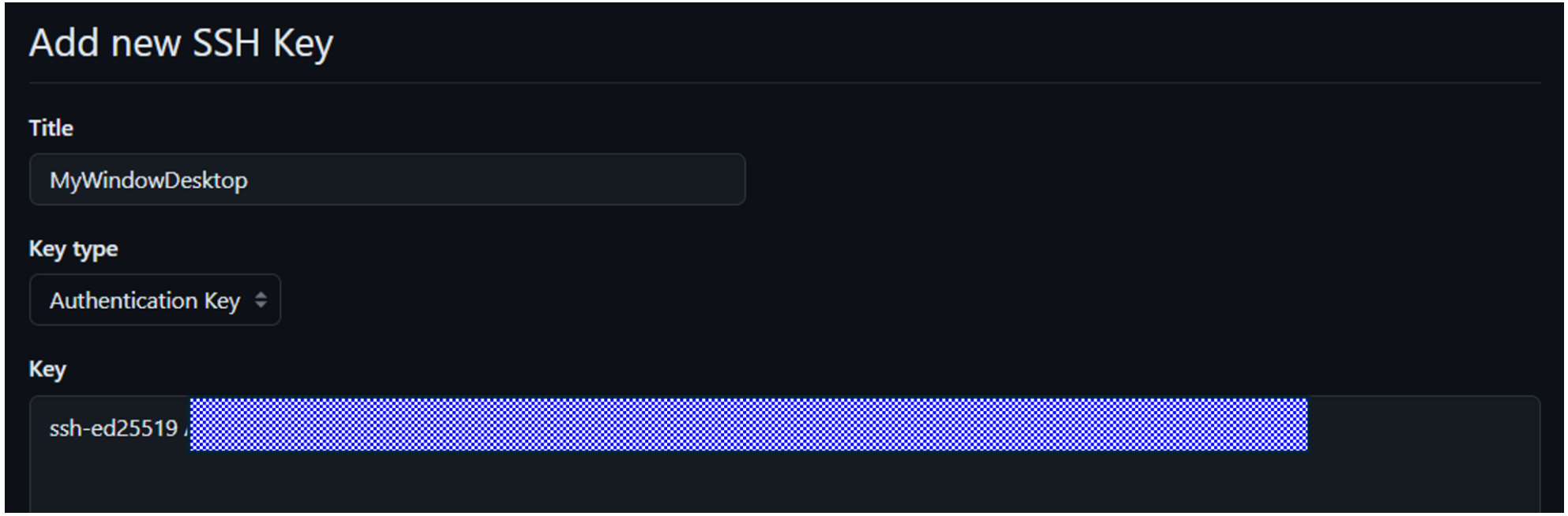
```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (master)
$ clip < /c/Users/jeong/.ssh/id_ed25519.pub
```

macOS $ pbcopy < ~/.ssh/id_ed25519.pub



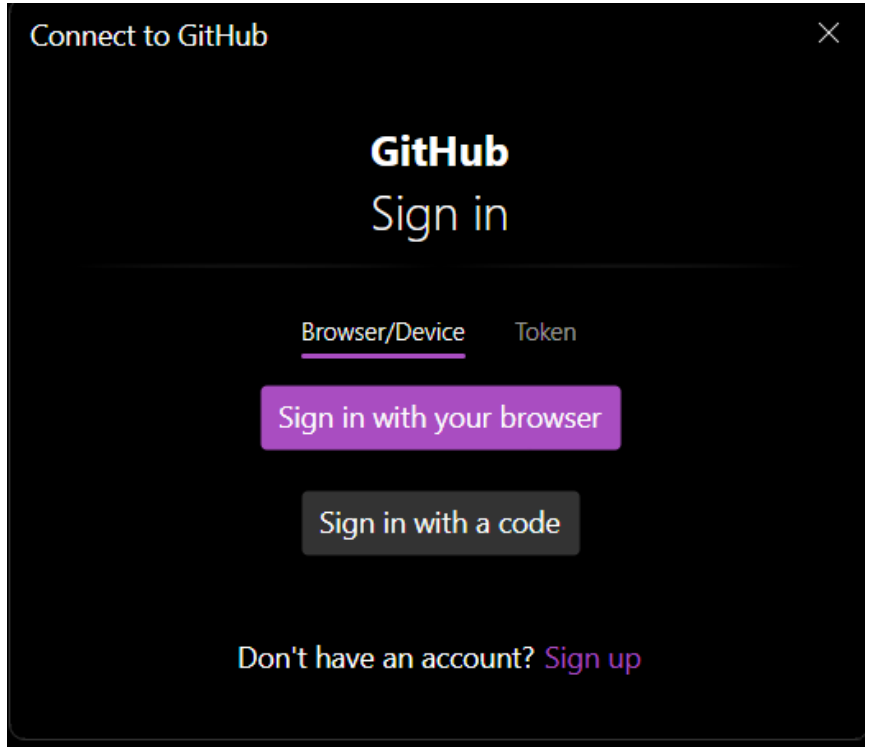


GitHub 레포 연결하기 - 내 깃헙 settings에 연결하기




```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (main)
$ git push origin master
```

GitHub 레포 연결하기




```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (main)
$ git push origin master
Everything up-to-date
```

Gitbash 1. 변경사항 새로 만들고 커밋, 푸시하기

- git add .
- git status
- git commit -m “커밋 메시지”
- git push origin master (git push origin main)

Gitbash 2. 레포지토리에 콜라보레이터 추가하기

- 레포지토리 이름 하단의 “settings” 메뉴 선택
- 새로 열린 메뉴 창 왼쪽에 Collaborators 선택
- Manager acces에 Add people 하여, 메일주소로 콜라보레이터 추가

Gitbash 3. 브랜치를 만들고 체크아웃하기

- git checkout -B [브랜치 이름]
- git checkout [원래 브랜치 이름]

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (main)
$ git checkout -B local
Switched to a new branch 'local'
```

```
jeong@swd47ag925 MINGW64 ~/Documents/GitTest/pushtest (local)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

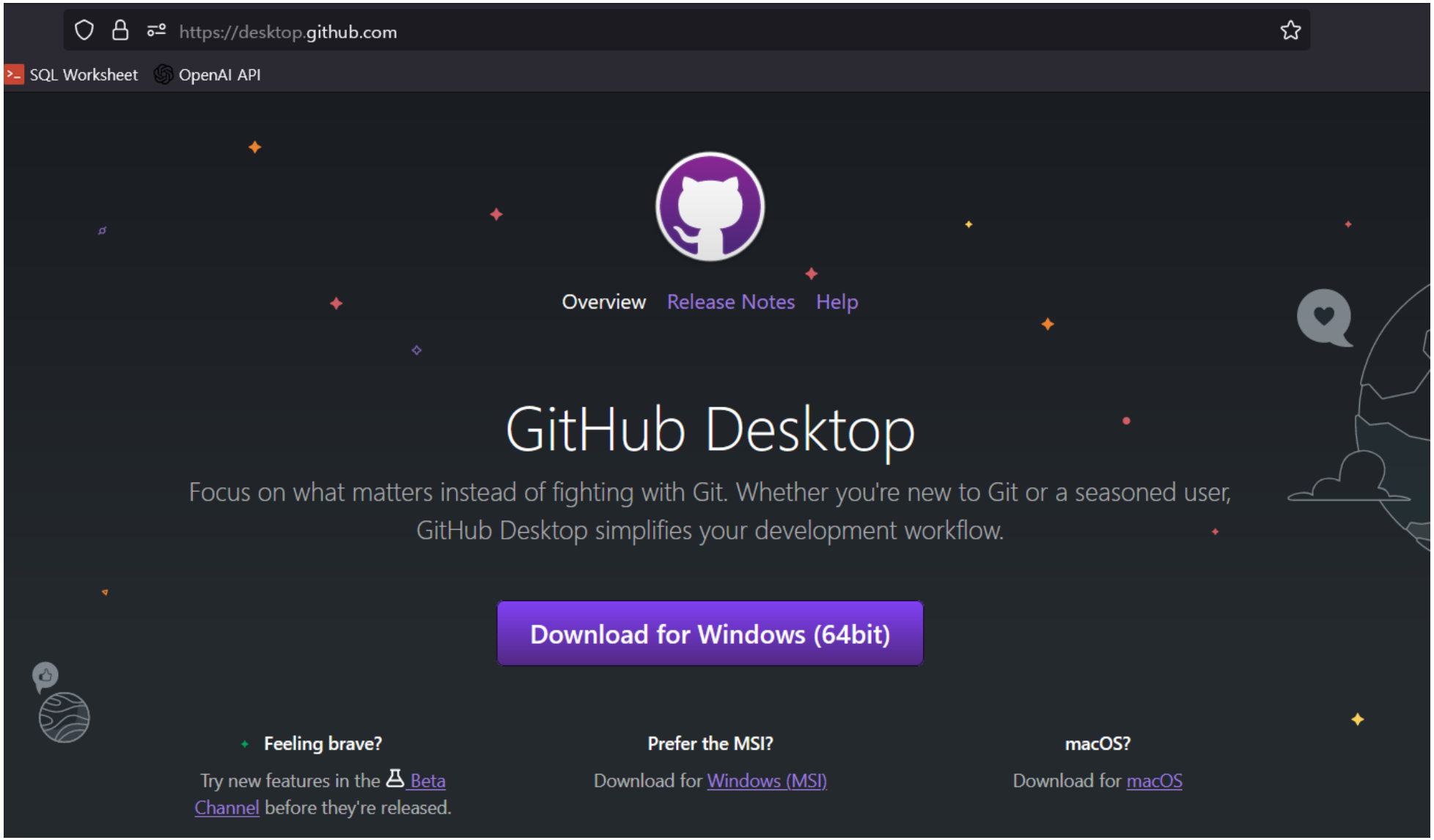
Gitbash 4. 새 디렉토리 만들고, 맘에 드는 공개 레포 클론하기

- cd [폴더 이름] : 폴더 이름 위치로 이동
- mkdir [새 폴더 이름] : [새 폴더 이름]을 가진 폴더 만들기
- git clone [깃레포 주소] : 공개된 깃 레포를 주소로 다운로드

Gitbash 5. 레포지토리 지우기

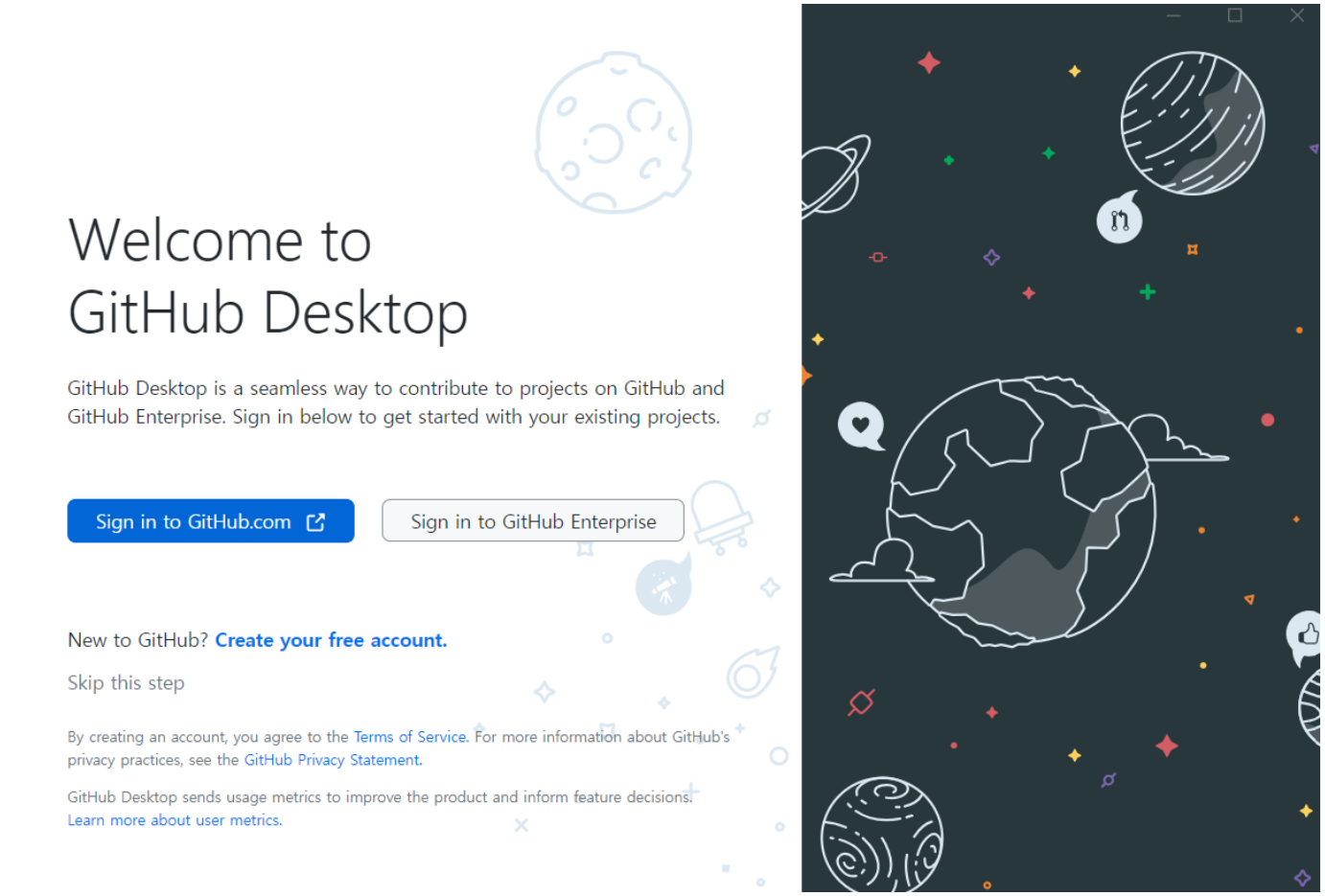
- 레포지토리 이름 하단의 “settings” 메뉴 선택
- 새로 열린 메뉴 창 왼쪽에 General 선택
- 제일 아래쪽 Danger Zone 확인,
- Delete this repository

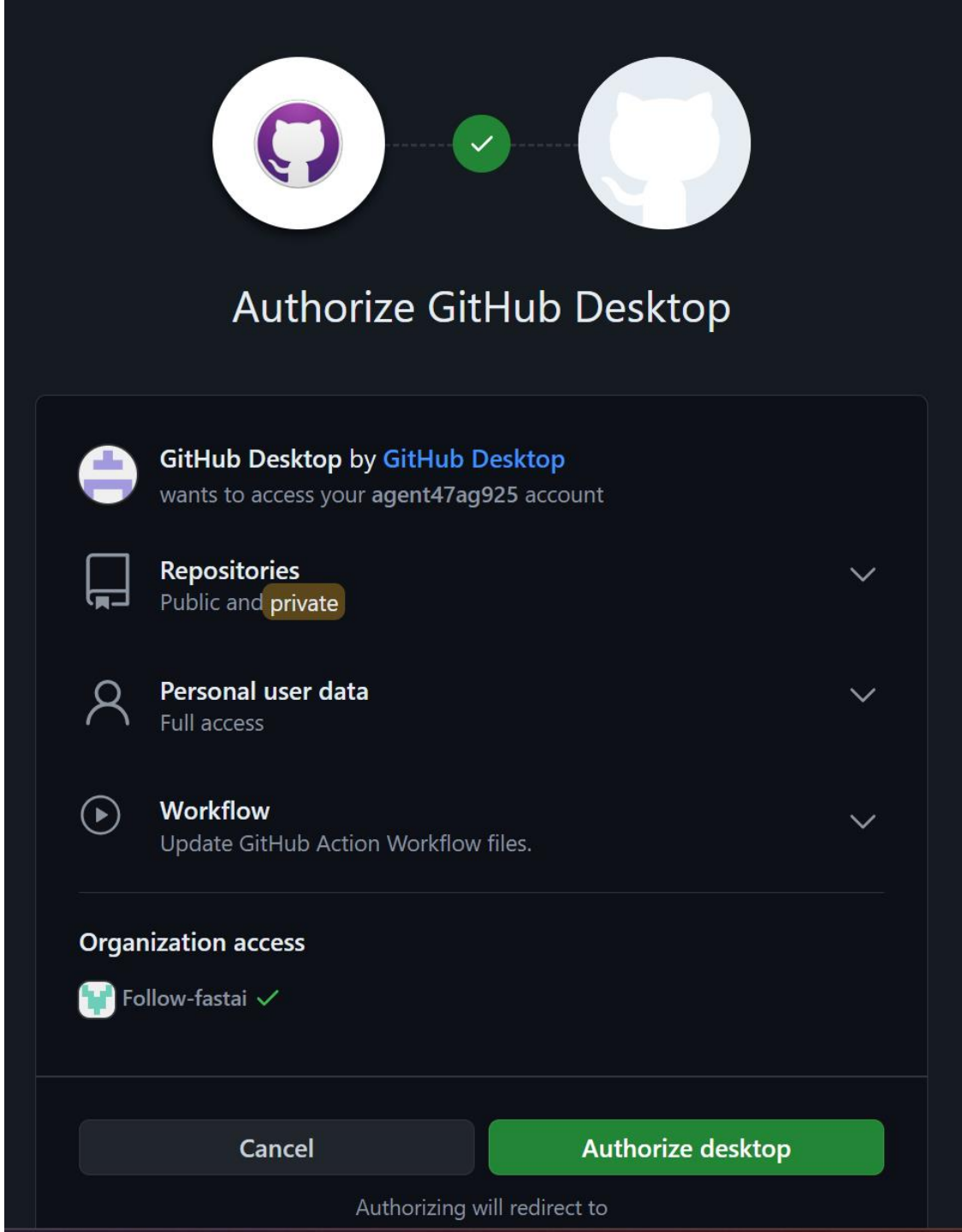
Git GUI – Github Desktop



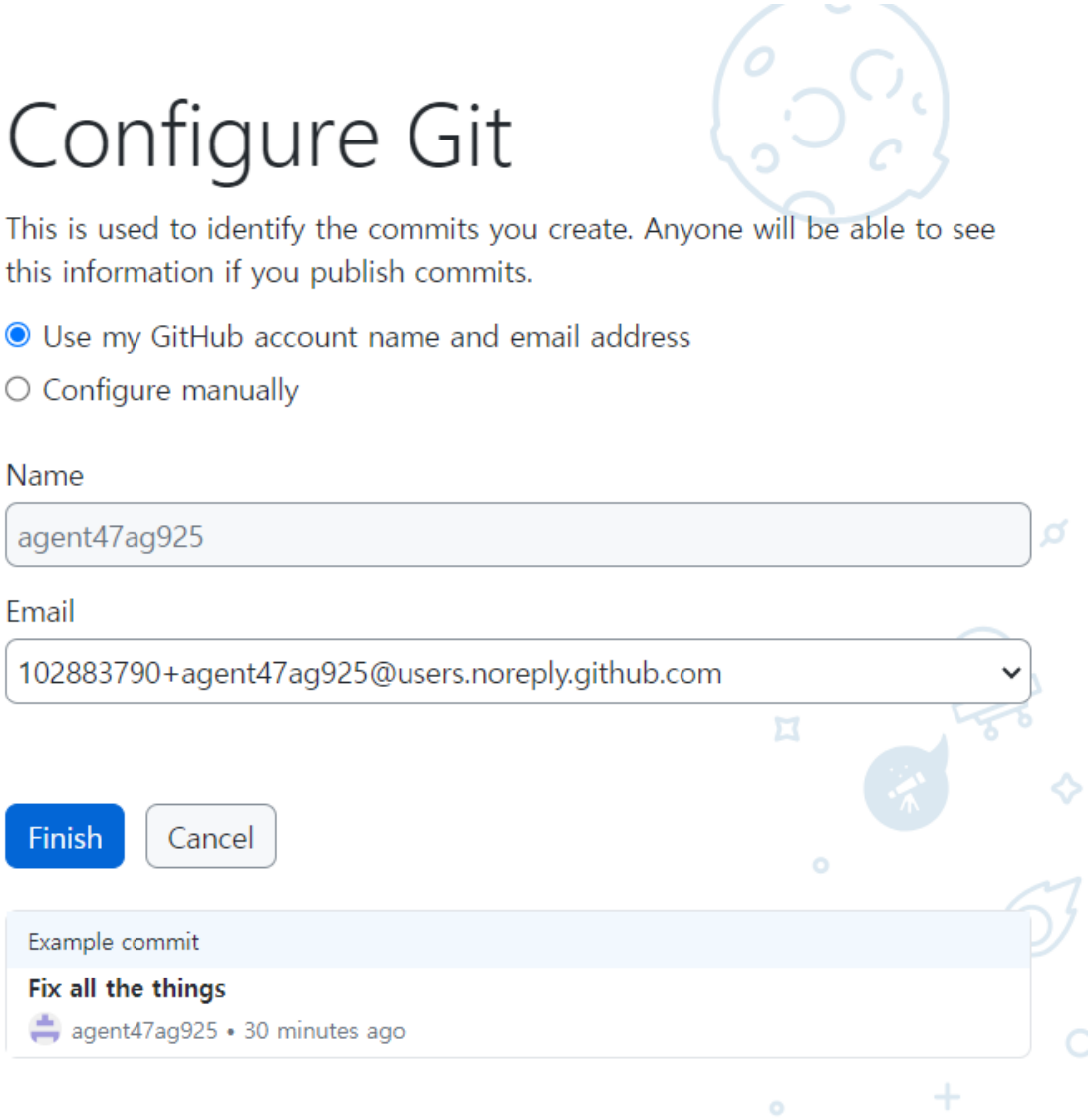
https://desktop.github.com/

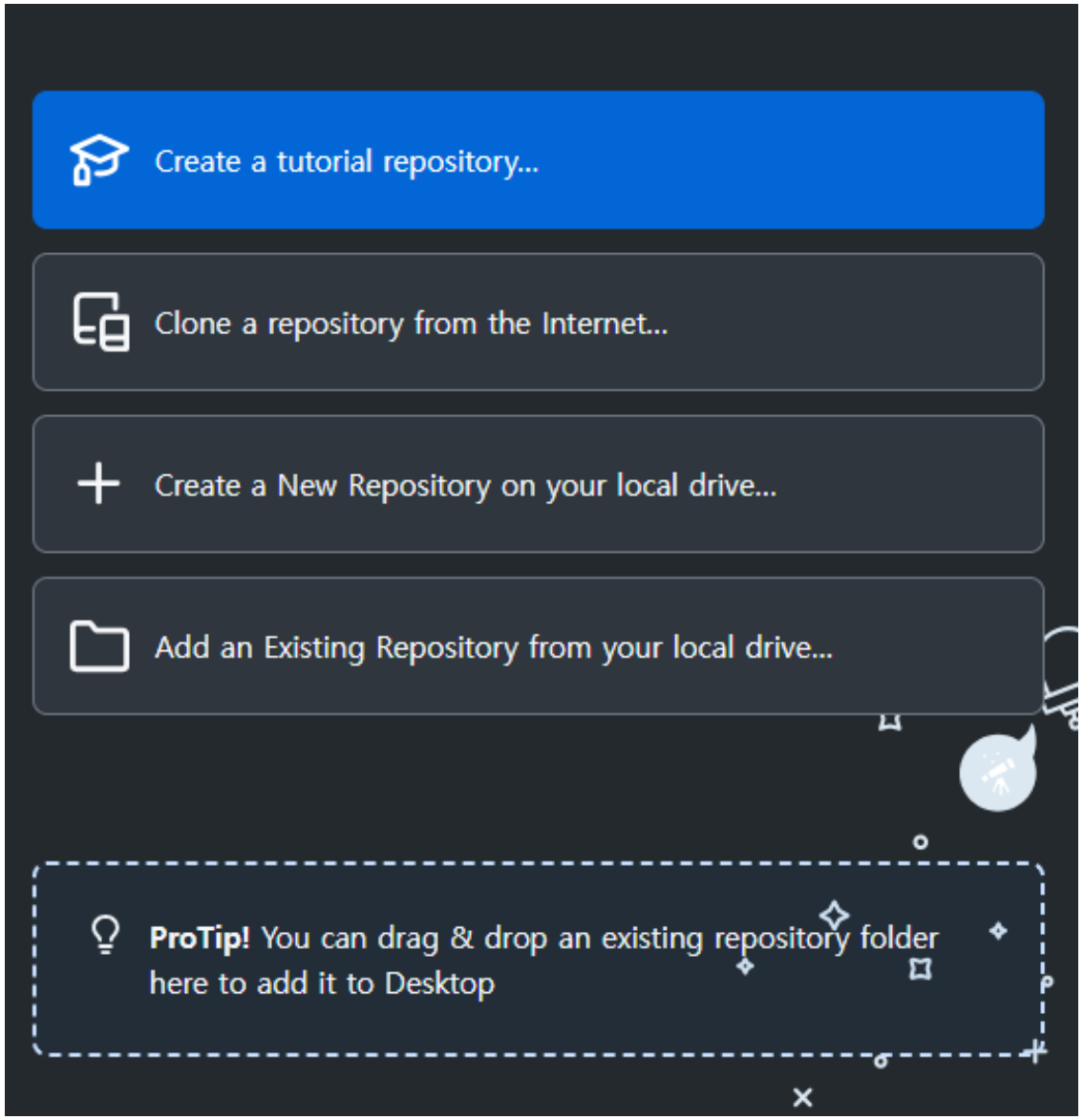
Git GUI - Github Desktop



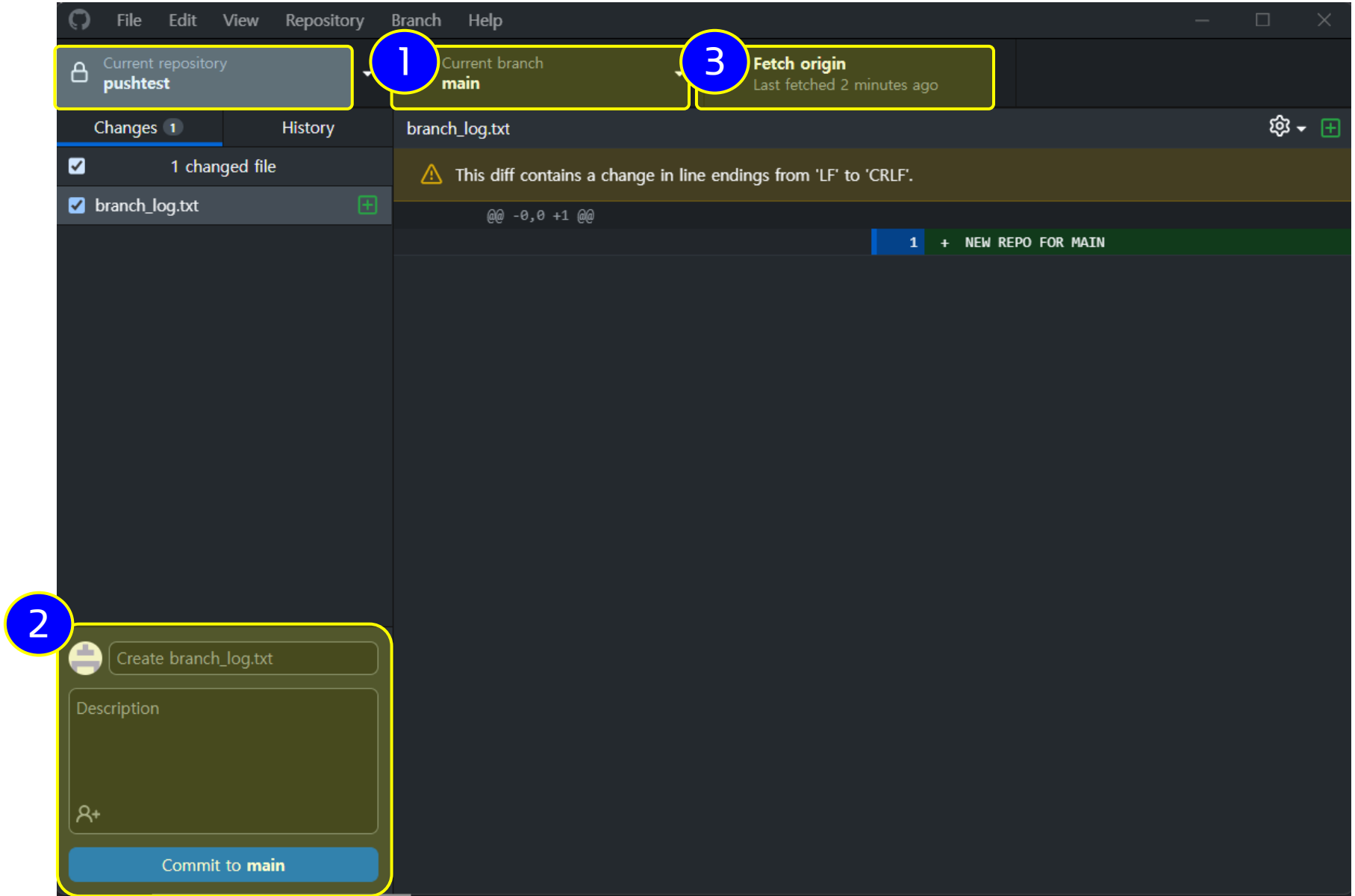


Git GUI - Github Desktop

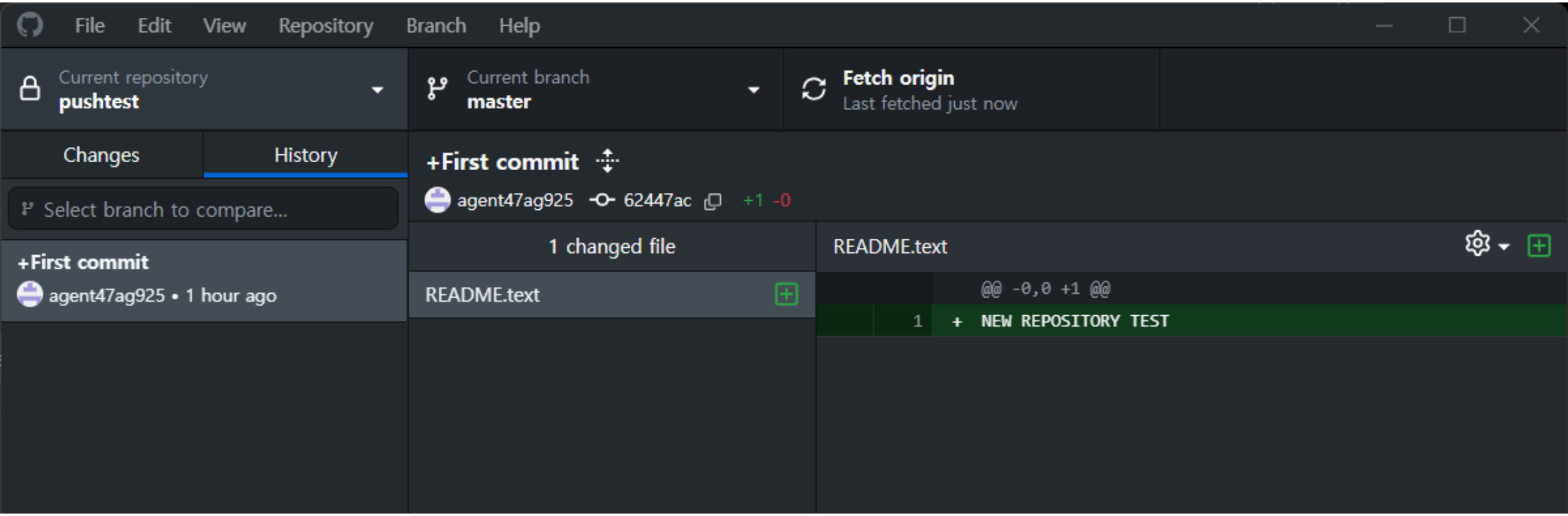
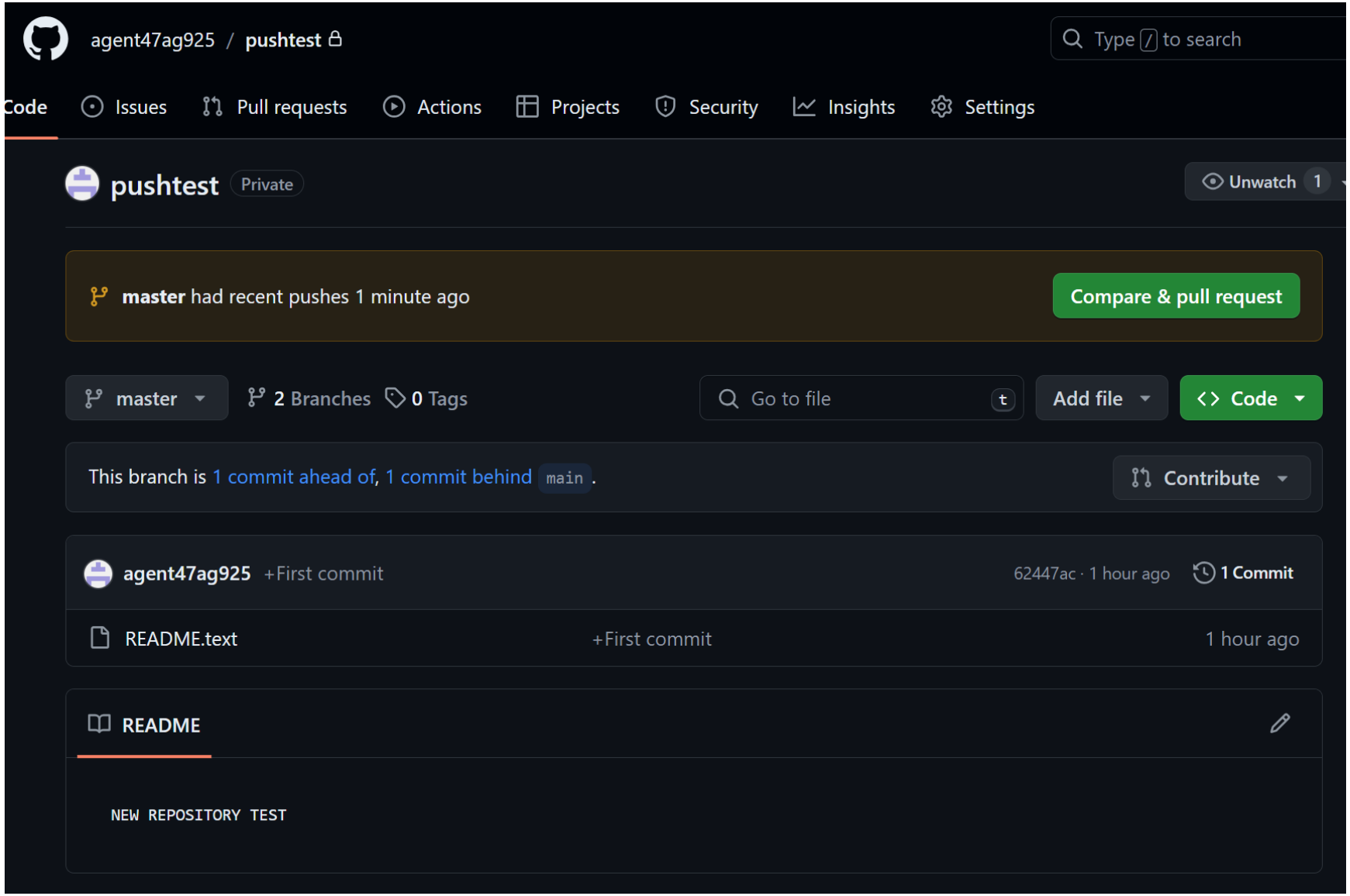




Git GUI - Github Desktop



Git GUI - Github Desktop



Git GUI – Github Desktop